Linear Programming modeling techniques

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Outline

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Linearization techniques
- Min-max and max-min objective functions
- Absolute values
- Ratios and percentages

What cannot be modeled with Linear Programming

From a problem description to an LP model

Introduction: From a problem description to an LP model

- So far the LP problems were given: in this chapter we learn how to **write** them, starting from the description of a decision problem
- Writing a model is the first step of every application of Linear Programming: the simplex algorithm can only solve the model we write
- We first fix a method in three steps and the conventions on the data, then we apply them to the classical models of production planning, blending, transportation and multi-period planning

What is a model

- A **model** of an optimization problem is a mathematical description of its feasible solutions and of the quantity to be optimized
- Writing a model requires three steps, always in this order

The three steps of a model

1. **Decision variables.** Identify the decisions to be taken and associate one variable with each of them. For every variable, state its unit of measure and its domain.

2. **Objective function.** Write, as a linear function of the decision variables, the quantity to be maximized or minimized.

3. **Constraints.** Write, as linear equalities or inequalities, every requirement that a feasible solution has to satisfy.

Symbolic input data

- The model is written with **symbolic input data**, so that it describes the problem for every instance, and not only for one of them

- We use the following conventions:
  1. the sizes of the instance are integer numbers, for example $r \in \mathbb{Z}_{\geq 1}$ and $s \in \mathbb{Z}_{\geq 1}$;
  2. the numerical data are rational numbers, for example $a_i \in \mathbb{Q}_{>0}$ and $w_{ij} \in \mathbb{Q}_{\geq 0}$;
  3. the index sets are written explicitly, for example $\forall i \in \{1, 2, \ldots, r\}$.

Linear Programming models

- A model is a **Linear Programming** model if:
  - its objective function and all its constraints are **linear**
  - all its variables are **continuous**

- In this case the model can be solved with the **simplex algorithm**

Before writing the first models, let us list the mistakes to avoid.

The three most frequent mistakes

The three most frequent mistakes when writing an LP model are:

1. forgetting the **non-negativity constraints** of the variables;
2. writing constraints which are not dimensionally consistent, that is, which sum quantities with different units of measure;
3. using a variable in the objective function without defining it among the decision variables.

Summary

Writing a model:

- Three steps, in this order: decision variables (with unit of measure and domain), objective function, constraints
- Symbolic input data: integer sizes, rational numerical data, explicit index sets

LP models:

- Linear objective function and constraints, continuous variables: solvable with the simplex algorithm
- Watch out for non-negativity, dimensional consistency and undefined variables

Production planning

Introduction: Production planning

- The first classical model: a company decides how much to produce of each product, with **limited resources**
- It is the simplest LP model, and its form is exactly the **canonical maximization form**
- We write the symbolic model, and then the model of a small instance with its optimal solution

The problem

- A company produces $r \in \mathbb{Z}_{\geq 1}$ different products using $s \in \mathbb{Z}_{\geq 1}$ limited resources
- For each product $j \in \{1, 2, \ldots, r\}$, the value $c_j \in \mathbb{Q}_{>0}$ is the profit of one unit of product
- For each resource $i \in \{1, 2, \ldots, s\}$, the value $b_i \in \mathbb{Q}_{>0}$ is the available quantity, and $a_{ij} \in \mathbb{Q}_{\geq 0}$ is the quantity of resource i consumed by one unit of product j
- The company aims to determine a **maximum profit** production plan

Decision variables

- We introduce the following r non-negative variables:

  $x_j =$ units of product j produced, $\qquad \forall j \in \{1, 2, \ldots, r\}$.

The production planning model

$$\max \quad \sum_{j=1}^{r} c_j \, x_j$$

$$\text{subject to} \quad \sum_{j=1}^{r} a_{ij} \, x_j \leq b_i, \quad \forall i \in \{1, 2, \ldots, s\},$$

$$x_j \geq 0, \quad \forall j \in \{1, 2, \ldots, r\}.$$

- The objective function is the **total profit**
- The constraints on the resources impose that the consumption of every resource does not exceed its availability
- This is an LP problem in **canonical maximization form**

Let us now write the model of an instance with two products and three resources.

Example: production planning (1/2)

Consider an instance with $r = 2$ products and $s = 3$ resources:

	$j = 1$	$j = 2$
c_j	3	5
a_{1j}	1	0
a_{2j}	0	2
a_{3j}	3	2

b_1	b_2	b_3
4	12	18

Example: production planning (2/2)

The LP model for this instance reads as follows:

$$
\begin{array}{rrcl}
\max & 3\,x_1 + 5\,x_2 & & \\
\text{subject to} & x_1 & \leq & 4 \\
& 2\,x_2 & \leq & 12 \\
& 3\,x_1 + 2\,x_2 & \leq & 18 \\
& x_1, \quad x_2 & \geq & 0
\end{array}
$$

- An optimal solution is $\tilde{x}_1 = 2$ and $\tilde{x}_2 = 6$, with optimal value $z(\,LP\,) = 36$

Summary

Production planning:

- Variables x_j: units of product j produced, non-negative
- Maximize the total profit $\sum_j c_j x_j$
- One constraint per resource: consumption $\sum_j a_{ij} x_j$ at most the availability b_i

Form:

- The model is an LP problem in canonical maximization form
- In the instance: $\tilde{x}_1 = 2$, $\tilde{x}_2 = 6$, $z(LP) = 36$

Blending

Introduction: Blending

- A final product is obtained by **mixing** raw materials, at minimum cost, with requirements on its composition
- Blending models appear for animal feeds, fuels, alloys, food products
- Here we also meet the first **linearization**: a requirement on a fraction of the blend, which looks non-linear, is rewritten as a linear constraint

The problem

- A quantity $d \in \mathbb{Q}_{>0}$ of a final product has to be obtained by mixing $r \in \mathbb{Z}_{\geq 1}$ raw materials
- For each raw material $j \in \{1, 2, \ldots, r\}$, the value $c_j \in \mathbb{Q}_{>0}$ is the cost of one unit
- The final product has to contain at least a quantity $q_k \in \mathbb{Q}_{\geq 0}$ of each of $s \in \mathbb{Z}_{\geq 1}$ components, and $p_{kj} \in \mathbb{Q}_{\geq 0}$ is the quantity of component k contained in one unit of raw material j
- The company aims to determine a **minimum cost** blend

Decision variables

- We introduce the following r non-negative variables:

$$x_j = \text{ units of raw material } j \text{ used in the blend,}$$

$$\forall j \in \{1, 2, \ldots, r\}.$$

The blending model

$$\min \quad \sum_{j=1}^{r} c_j \, x_j$$

$$\text{subject to} \quad \sum_{j=1}^{r} x_j = d,$$

$$\sum_{j=1}^{r} p_{kj} \, x_j \geq q_k, \quad \forall k \in \{1, 2, \ldots, s\},$$

$$x_j \geq 0, \qquad \forall j \in \{1, 2, \ldots, r\}.$$

- The first constraint imposes that the total quantity produced is d
- The constraints on the components impose the minimum quantity of each component

Fractions of the blend

- A requirement of the form "*the raw material j has to be at least a fraction α of the blend*" is not written as

$$\frac{x_j}{\sum_\ell x_\ell} \geq \alpha,$$

which is not a linear constraint

- It is written instead by multiplying both sides by the denominator, which is positive:

$$x_j \geq \alpha \sum_{\ell=1}^{r} x_\ell \qquad \Longleftrightarrow \qquad (1-\alpha)\, x_j - \alpha \sum_{\ell \neq j} x_\ell \geq 0.$$

Let us now write the model of an animal feed blend, with a requirement on the fraction of one raw material.

Example: blending (1/3)

- A company has to produce $d = 100$ kilograms of animal feed by mixing $r = 3$ raw materials
- The costs per kilogram are $c_1 = 0.80$, $c_2 = 0.50$ and $c_3 = 0.30$ euros
- The feed has to contain at least $q_1 = 8$ kilograms of proteins, and one kilogram of each raw material contains, respectively, $p_{11} = 0.12$, $p_{12} = 0.08$ and $p_{13} = 0.03$ kilograms of proteins
- Moreover, the first raw material has to be at least the 20% of the blend

Example: blending (2/3)

The LP model for this instance reads as follows:

$$
\begin{aligned}
\min \quad & 0.80\,x_1 + 0.50\,x_2 + 0.30\,x_3 & & \\
\text{subject to} \quad & x_1 + x_2 + x_3 & = & \ 100 \\
& 0.12\,x_1 + 0.08\,x_2 + 0.03\,x_3 & \geq & \ 8 \\
& 0.80\,x_1 - 0.20\,x_2 - 0.20\,x_3 & \geq & \ 0 \\
& x_1, \quad x_2, \quad x_3 & \geq & \ 0
\end{aligned}
$$

- The third constraint is the proportion requirement $x_1 \geq 0.20\,(x_1 + x_2 + x_3)$, rewritten as a linear constraint

Example: blending (3/3)

- An optimal solution is

$$\tilde{x}_1 = 20, \qquad \tilde{x}_2 = 64, \qquad \tilde{x}_3 = 16,$$

with optimal value

$$z(\,LP\,) = 0.80 \cdot 20 + 0.50 \cdot 64 + 0.30 \cdot 16 = 52.80 \text{ euros.}$$

Summary

Blending:

- Variables x_j: units of raw material j used in the blend, non-negative
- Minimize the total cost; the total quantity is equal to d; at least q_k of each component

Fractions of the blend:

- "At least a fraction α" is not written as a ratio: multiply by the positive denominator, $x_j \geq \alpha \sum_\ell x_\ell$
- In the instance: $\tilde{x} = (20, 64, 16)$, $z(\,LP\,) = 52.80$ euros

Transportation

Introduction: Transportation

- A product stored in several **warehouses** has to reach several **clients**, at minimum transportation cost
- The variables now have **two indices**: one for the warehouse and one for the client
- We write the model, see when it is feasible, and solve a small instance

The problem

- A company has $s \in \mathbb{Z}_{\geq 1}$ warehouses and has to serve $r \in \mathbb{Z}_{\geq 1}$ clients
- For each warehouse $i \in \{1, 2, \ldots, s\}$, the value $a_i \in \mathbb{Q}_{>0}$ is the available quantity of product
- For each client $j \in \{1, 2, \ldots, r\}$, the value $d_j \in \mathbb{Q}_{>0}$ is the demand, and $c_{ij} \in \mathbb{Q}_{\geq 0}$ is the cost of transporting one unit of product from warehouse i to client j
- The company aims to determine a **minimum cost** transportation plan

Decision variables

- We introduce the following $s \cdot r$ non-negative variables:

$$x_{ij} = \text{ units transported from warehouse } i \text{ to client } j,$$

$$\forall i \in \{1, 2, \ldots, s\}, \ \forall j \in \{1, 2, \ldots, r\}.$$

The transportation model

$$\min \quad \sum_{i=1}^{s} \sum_{j=1}^{r} c_{ij}\, x_{ij}$$

$$\text{subject to} \quad \sum_{j=1}^{r} x_{ij} \leq a_i, \quad \forall i \in \{1, 2, \ldots, s\},$$

$$\sum_{i=1}^{s} x_{ij} \geq d_j, \quad \forall j \in \{1, 2, \ldots, r\},$$

$$x_{ij} \geq 0, \quad \forall i \in \{1, 2, \ldots, s\},\ \forall j \in \{1, 2, \ldots, r\}.$$

Reading the transportation model

- The availability constraints impose that no warehouse ships more than its availability
- The demand constraints impose that every client receives at least its demand
- The model is feasible if and only if $\sum_i a_i \geq \sum_j d_j$

Let us now solve an instance
with two warehouses and two clients.

Example: transportation (1/2)

- Consider an instance with $s = 2$ warehouses, of availabilities $a_1 = 50$ and $a_2 = 60$
- and $r = 2$ clients, of demands $d_1 = 30$ and $d_2 = 80$
- The transportation costs are:

c_{ij}	$j = 1$	$j = 2$
$i = 1$	4	6
$i = 2$	5	3

Example: transportation (2/2)

- An optimal solution is

$$\tilde{x}_{11} = 30, \qquad \tilde{x}_{12} = 20, \qquad \tilde{x}_{21} = 0, \qquad \tilde{x}_{22} = 60,$$

with optimal value

$$z(\,LP\,) = 4 \cdot 30 + 6 \cdot 20 + 5 \cdot 0 + 3 \cdot 60 = 420.$$

Summary

Transportation:

- Variables x_{ij} with two indices: units transported from warehouse i to client j
- Minimize the total transportation cost
- No warehouse ships more than a_i; every client receives at least d_j

Feasibility:

- The model is feasible if and only if $\sum_i a_i \geq \sum_j d_j$
- In the instance: $z(\,LP\,) = 420$

Multi-period planning

Introduction: Multi-period planning

- The decisions are now taken over a **time horizon**: what is produced in one period can be stored and sold later
- We need two families of variables: production and storage
- The key ingredient is the **inventory balance constraint**, which links consecutive periods

The problem

- A company has to satisfy the demand of a product over a time horizon of $n \in \mathbb{Z}_{\geq 1}$ periods
- For each period $t \in \{1, 2, \ldots, n\}$:
  - $d_t \in \mathbb{Q}_{\geq 0}$ is the demand
  - $p_t \in \mathbb{Q}_{> 0}$ is the maximum production
  - $c_t \in \mathbb{Q}_{> 0}$ is the unit production cost
  - $h_t \in \mathbb{Q}_{> 0}$ is the unit cost of storing one unit from period t to period $t + 1$

Decision variables

- We introduce the following $2\,n$ non-negative variables:

$$\begin{cases} x_t = & \text{units produced in period } t \\ y_t = & \text{units stored at the end of period } t \end{cases}$$

$$\forall t \in \{1, 2, \ldots, n\}.$$

The multi-period model

$$\min \quad \sum_{t=1}^{n} c_t \, x_t + \sum_{t=1}^{n} h_t \, y_t$$

$$\begin{aligned}
\text{subject to} \quad & y_{t-1} + x_t - y_t = d_t, \quad \forall t \in \{1, 2, \ldots, n\}, \\
& x_t \leq p_t, \quad \forall t \in \{1, 2, \ldots, n\}, \\
& x_t \geq 0, \quad \forall t \in \{1, 2, \ldots, n\}, \\
& y_t \geq 0, \quad \forall t \in \{1, 2, \ldots, n\},
\end{aligned}$$

with $y_0 = 0$.

Inventory balance constraints

- The constraints $y_{t-1} + x_t - y_t = d_t$ are the **inventory balance constraints**
- In every period, what was in stock at the end of the previous period plus what is produced equals the demand plus what is left in stock:

$$\underbrace{y_{t-1}}_{\text{stock in}} + \underbrace{x_t}_{\text{produced}} = \underbrace{d_t}_{\text{demand}} + \underbrace{y_t}_{\text{stock out}}$$

- They are the key of every multi-period model, and they **link consecutive periods**

Summary

Multi-period planning:

- Two families of variables: x_t, units produced in period t, and y_t, units stored at the end of period t, with $y_0 = 0$
- Minimize the production cost plus the storage cost
- The production in each period is at most p_t

Inventory balance:

- $y_{t-1} + x_t - y_t = d_t$ for every period: the constraints that link consecutive periods

Linearization techniques

Introduction: Linearization techniques

- Some requirements are **not linear** as they are written, but they can be rewritten as linear constraints, possibly with the help of additional variables
- In this way problems that look non-linear can still be solved with the simplex algorithm
- We present the three most frequent cases:
  1. min-max and max-min objective functions
  2. absolute values
  3. ratios and percentages

Min-max objective functions

An objective function of the form

$$\min \quad \max_{i \in \{1,2,\ldots,m\}} \left\{ \sum_{j=1}^{n} a_{ij}\, x_j \right\}$$

is linearized by introducing one additional free variable w and writing

$$\min \; w$$

$$\text{subject to} \quad w \geq \sum_{j=1}^{n} a_{ij}\, x_j, \quad \forall i \in \{1, 2, \ldots, m\}.$$

Why the linearization works

- The m constraints impose that w is not smaller than each of the m linear functions, that is, that w is not smaller than their maximum
- Since w is minimized, at an optimal solution it is exactly equal to that maximum
- Symmetrically, a **max-min** objective function is linearized by introducing one additional free variable w, maximizing it and imposing $w \leq \sum_j a_{ij} x_j$ for every i
- The linearization works only in these two directions: a maximization of a maximum, or a minimization of a minimum, cannot be linearized in this way

Absolute values

A term $\left|\sum_{j=1}^{n} a_{ij}\, x_j\right|$ appearing with a positive coefficient in a minimized objective function is linearized by introducing one additional non-negative variable w_i and writing

$$w_i \geq \sum_{j=1}^{n} a_{ij}\, x_j \qquad \text{and} \qquad w_i \geq -\sum_{j=1}^{n} a_{ij}\, x_j .$$

- The two constraints impose $w_i \geq \left|\sum_j a_{ij}\, x_j\right|$, and the minimization makes the inequality tight at an optimal solution

Absolute values: an alternative linearization

- Under the same condition (positive coefficient in a minimized objective function, or negative coefficient in a maximized one), the same term is linearized by writing

$$\sum_j a_{ij}\, x_j = w_i^+ - w_i^-, \qquad w_i^+, w_i^- \geq 0,$$

and by replacing the absolute value with $w_i^+ + w_i^-$

- The optimization makes at least one of w_i^+ and w_i^- equal to 0 at an optimal solution, and then

$$w_i^+ + w_i^- = \left| \sum_j a_{ij}\, x_j \right|$$

- Without that condition, both variables could be positive and the model would be wrong

Ratios and percentages

A constraint of the form

$$\frac{\sum_{j=1}^{n} a_j\, x_j}{\sum_{j=1}^{n} e_j\, x_j} \leq \gamma, \qquad \text{with} \qquad \sum_{j=1}^{n} e_j\, x_j > 0,$$

is linearized by multiplying both sides by the denominator, which is positive:

$$\sum_{j=1}^{n} \left(a_j - \gamma\, e_j\right) x_j \leq 0.$$

- The same technique applies to the percentage requirements of the blending models

When the technique is correct

- The technique is correct only when the denominator is known to be positive for every feasible solution
- Otherwise multiplying by it would change the direction of the inequality

Summary

Min-max and max-min:

- One additional free variable w: $\min w$ with $w \geq \sum_j a_{ij} x_j$ for every i (symmetrically for max-min); only in these two directions

Absolute values:

- With a positive coefficient in a minimized objective function: $w_i \geq \pm \sum_j a_{ij} x_j$, or $\sum_j a_{ij} x_j = w_i^+ - w_i^-$

Ratios and percentages:

- Multiply by the denominator, only if it is positive for every feasible solution

What cannot be modeled with Linear Programming

Introduction: What cannot be modeled with Linear Programming

- Linear Programming has continuous variables and linear constraints: some requirements are out of its reach
- Recognizing them is part of the modeling: they need **integer** variables
- Even then, the LP model obtained by ignoring them remains useful, as a **relaxation**

Requirements that need integer variables (1/2)

The following requirements cannot be written as linear constraints on continuous variables, and they need **integer** variables:

1. **indivisible quantities**, for example the number of machines to buy or of vehicles to use;

2. **yes/no decisions**, for example whether to open a warehouse or not;

3. **fixed costs**, that is, a cost which is paid only if the associated activity is performed.

Requirements that need integer variables (2/2)

4. **either-or constraints**, that is, requirements of which at least one out of two has to hold;

5. **minimum positive quantities**, for example "*either produce nothing, or produce at least q units*".

These requirements are the subject of **Integer Linear Programming**.

A frequent mistake

- A frequent mistake is to write a fixed cost as a non-linear term of the objective function
- or to impose a minimum positive quantity with the constraint $x_j \geq q$: the latter forbids the solution $x_j = 0$, which instead has to remain feasible

The LP relaxation

- Whenever the requirements of a problem are of the kind listed above, the LP model obtained by ignoring them is still useful: it is a **relaxation** of the correct model

- Its optimal value is:
  - an **upper bound** on the optimal value of the correct model, for a maximization problem
  - a **lower bound**, for a minimization problem

Summary

Beyond Linear Programming:

- Indivisible quantities, yes/no decisions, fixed costs, either-or constraints and minimum positive quantities need integer variables: Integer Linear Programming
- $x_j \geq q$ is not a minimum positive quantity: it forbids $x_j = 0$

Relaxation:

- The LP model without these requirements gives an upper bound (maximization) or a lower bound (minimization) on the optimal value of the correct model

Summary of the chapter (1/3)

Writing an LP model:

- Decision variables (unit of measure and domain), objective function, constraints, with symbolic input data
- Avoid the three frequent mistakes: non-negativity, dimensional consistency, undefined variables

Summary of the chapter (2/3)

Four classical models:

- Production planning: maximum profit with limited resources, in canonical maximization form
- Blending: minimum cost blend of quantity d with minimum quantities of the components
- Transportation: minimum cost plan from warehouses to clients, feasible iff $\sum_i a_i \geq \sum_j d_j$
- Multi-period planning: production and storage, linked by the inventory balance constraints

Summary of the chapter (3/3)

Linearization techniques:

- Min-max and max-min objective functions: one additional free variable w
- Absolute values: $w_i \geq \pm \sum_j a_{ij} x_j$, or $w_i^+ - w_i^-$, under the right sign condition
- Ratios and percentages: multiply by the denominator, if it is positive

Limits of Linear Programming:

- Indivisible quantities, yes/no decisions, fixed costs, either-or constraints and minimum positive quantities need Integer Linear Programming
- Ignoring them gives a relaxation, whose optimal value is a bound